STAT 112 Introduction to Data Processing and Visualization Final Project

Department of Statistics
Middle East Technical University

Arda Batın Aydın
Ankara, Turkey
arda.aydin_01@metu.edu.tr

Kadir Emre Gülen
Ankara, Turkey
gulen.kadir@metu.edu.tr

Necdet Kerem Özaydın
Ankara, Turkey
kerem.ozaydin@metu.edu.tr

Göktuğ Arık
Ankara, Turkey
goktug.arik@metu.edu.tr

Prepared for: Prof. Dr. Ceylan Yozgatlıgil
Date: 16.01.2026

Strategic E-Commerce Analysis

Arda Batın Aydın
First Research Question
Department of Statistics
Middle East Technical University
Ankara, Türkiye
arda.aydin_01@metu.edu.tr

Kadir Emre Gulen
Second Research Question
Department of Statistics
Middle East Technical University
Ankara, Türkiye
gulen.kadir@metu.edu.tr

Necdet Kerem Özaydın
Third Research Question
Department of Statistics
Middle East Technical University
Ankara, Türkiye
kerem.ozaydin@metu.edu.tr

Göktuğ Arık
Fourth Research Question
Department of Statistics
Middle East Technical University
Ankara, Türkiye
goktug.arik@metu.edu.tr

I.) Abstract

This study analyzes e-commerce customer behavior and purchasing patterns across different cities in Turkey. How transaction metrics relate to factors like age, membership levels, and gender were the primary factors that have been focused on. By examining 1,725 data points, it was determined by our team that understanding general consumer trends and finding useful insights for marketing and operational improvements were the aim for this study. The research covers various correlations in online shopping through detailed data cleaning, including fixing missing values, finding outliers, and organizing data types for multivariate visualizations. Specifically, the main concerns of this study were how basket values change across different categories and how gender factors affect it, how each membership tier spends and their spread among genders, the satisfaction scores of each category and the correlation between spending behavior, and finally, how discount rates affect each membership tier. The findings show clear links between loyalty programs, spending behavior among genders, and regional e-commerce trends in Turkey. By analyzing the results that have been deduced, the company and people that are interested in this field can use and benefit from it.

II.) Introduction

This report will investigate a detailed online dataset on e-commerce that takes into account several online purchasing metrics for several cities in Turkey. The objective of this study is primarily to understand the correlations between purchasing behaviors, discount rates and customer demographic information such as sex and customer membership levels. By finding correlations in those areas, some trends will be deduced, and the company of this dataset will benefit from focusing on the areas where they can implement efficient techniques to improve.

Through this analysis, some methods have been used to identify key trends which may assist businesses to strengthen their marketing plans. In this research, interrelation between a variety of variables was examined using Python tools and libraries such as Seaborn and MatplotLib for visualization. Also, a strict cleaning procedure of the provided data as the checklist given was ensured. Through answering specific research questions, conclusions were made regarding the impact of key variables upon the online shopping market of Turkey. Firstly, the key findings can be stated as the males in this data set are the ones who made more spending than females, surprisingly. Secondly, the gender factor does not affect spending among different memberships while memberships affect average spending seriously. Thirdly, it can be observed that this company is applying a strategy that benefits gold/silver tiers by providing more consistent and high value discount opportunities. Fourthly, home and kitchen category remains as the leading 'star' category, groceries is the actual one which need to be focused on as a field due to its low satisfaction score. In this report, Section 3 discusses data cleaning, and while section 4 emphasizes exploratory data analysis (EDA), section 5 concludes the report.

III.) Data Tidying and Cleaning Steps

a.) Getting Know the Data

Before getting into the details of data cleaning data types of each variable can be analyzed to understand what kind of data will be handled later. Therefore, an excel file was created to fulfill this requirement.

Variable Name	Description
Year (integer)	Calendar year of the order. Range: 2018–2024.
Order ID (string)	Unique order identifier (synthetic).
Gender (object)	Customer gender. Values: Female, Male.
Age (integer)	Customer age in years. Range: 18–70.
City (categorical)	Customer city in Türkiye (synthetic assignment). Example values include İstanbul, Ankara, İzmir, etc.
Membership Tier (categorical)	Loyalty/membership level. Values: Guest, Bronze, Silver, Gold.
Device Type (categorical)	Device used during purchase. Values: Mobile, Desktop, Tablet.
Traffic Source (categorical)	Main acquisition channel for the session. Values: Organic, Paid Search, Social, Email, Referral.
Payment Method (categorical)	Payment type. Values: Credit Card, Debit Card, Cash on Delivery, Bank Transfer, Digital Wallet.
Product Category (categorical)	Primary product category in the basket. Values: Electronics, Fashion, Home & Kitchen, Beauty, Sports, Books & Stationery, Groceries.
Items in Cart (integer)	Number of items purchased in the order. Range: 1–12.
Basket Value TRY (numeric)	Final basket value in Turkish Lira (TRY) after discounts.
Discount Rate (numeric)	Discount rate applied to the order (0–0.45)
Delivery Days (integer)	Delivery duration in days. Range: 1–10.
Satisfaction Score (numeric)	Post-delivery customer satisfaction score on a 1.0–5.0 scale (one decimal).
Repeat Purchase (bool)	Target variable indicating whether the customer made another purchase within 90 days.
	Values: 1 = Yes, 0 = No

By looking at these data datatypes an efficient approach has been adopted by our team and four research questions have been determined. Some variables were dropped due to the fact that they were unnecessary and causing untidiness in the data since they will not be taken into consideration.

b.) Data Cleaning Steps

The raw dataset was initially examined to understand its structure, variables and data datatypes (Checklist 1). The dataset consists of 1725 rows and 16 columns, with varied data

types such as float, integers and strings. Structure of this dataset was verified in order to prevent potential inconsistencies. No separation problems were seen, but several NaN values were detected (Checklist 2). Each column and row were inspected to verify they are adhering to the tidy data principles (Checklist 3). Since there were no untidy format detected while inspection, this dataset was considered as already a tidy one. Therefore, the application of data tidying techniques (stack/unstack, melt, pivot) were considered unnecessary. When the names of columns are examined, several typos (white spaces, upper-lower cases, no spaces etc.) were detected and were fixed using pandas (Checklist 4). Since this dataset is aimed at understanding customer behaviors, ORDERID column was found to be unnecessary; therefore, it was dropped (Checklist 5). No duplicates were found in this dataset; thus, no action is needed (Checklist 6). These initial data cleaning and tidying steps might be considered crucial as they are making the data set cleaner and much tidier; thus, it facilitates how the data analyzer processes it.

Unnecessary strings (such as % and * signs) and white spaces were found in some columns, and they were all removed using pandas (Checklist 7-8). This step facilitated how the analyzing was done since it affects viewing the dataset. Having noticed there is no standardization among string values, all strings were corrected and converted to lower-case strings (Checklist 9). Also, some non-unique observation cases (such as f3male and female) detected, and they were fixed to achieve uniqueness (Checklist 10). Since there were no day and month columns, creating a date column found to be unnecessary (Checklist 11). By using these steps, uniqueness and standardization were achieved among the data set. This affected how data was examined and was viewed.

Data types of each variable were checked again to be ensure of their data types since during the visualization part some calculations will be needed, and it is important to take data types into consideration (Checklist 12). Descriptive statistics of the data were examined to be sure there is no unusual behavior. Other than NaN values (as they will be handled in Step 16) there were no unusual behaviors found. (Checklist 13). As all the observations are within their domain, no action is needed. By using IQR formula, potential outliers were checked and found ones replaced with the mean (Checklist 14). All variables are analyzed again to be sure whether uniformity is achieved among all observations to adhere uniformity rule (Checklist 15). Finally, since all missing value ratios are low, dropping the columns were not taken into consideration. Therefore, all NaN values are filled with mean or mode depending on their data types (Checklist 16). If the variable is categorical, NaN values were filled with mode, and if the variable is numerical, NaN values were filled with mean. Having completed data cleaning part, this new clean data file was saved as a new csv file.

IV.) Exploratory Data Analysis (EDA)

A.) Data Overview

To interpret these data has been cleaned, further research and visualization techniques must be applied. Therefore, four research questions has been asked and analyzed to get answers in order to achieve the aim of our team.

B.) Research Questions

B.1.) Which category has the highest average basket value, and how does it differ among genders?

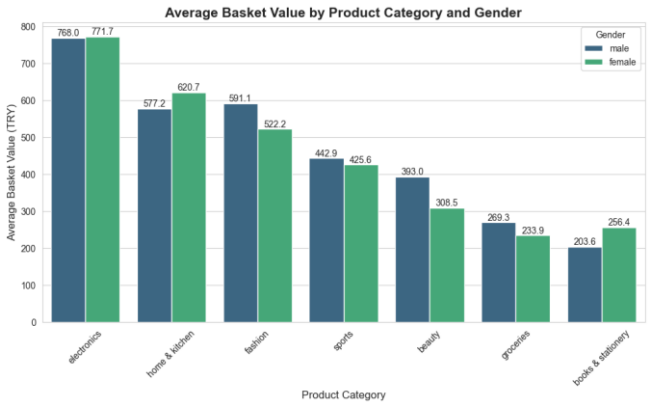

Figure 1: Bar Chart of Average Basket Value by Categories

Fig. 1 illustrates average basket values across different product categories, segmented by gender. Average basket value between 0 and 800 TRY of each category is represented by bars, and each gender is represented by colors in this graph, where blue represents the males, and green represents the females. This chart shows that the electronics category has the highest basket value, with negligible difference between males and females. The "Home & Kitchen" category has the second highest basket value; however, unlike electronics, it has a notable gender gap, which is nearly 45 TRY. It is also important to note that highest average basket value does not mean the highest spending. For example, in "Beauty" category, there is an interesting insight this graph shows that when the category is examined, males' spending can be seen much higher than females (nearly 27% higher). This might be indicating that while females shop more frequently, males might have higher transaction values per purchase in this category. Conversely, on "Books & Stationery" category, which has the lowest spending among all categories, females' spending is nearly 25% more per basket compared to males. This might be interpreted as females might have a stronger willingness to pay in reading and stationery categories.

B.2) The Membership Divide: How does Gender impact spending behavior within each Loyalty Tier?

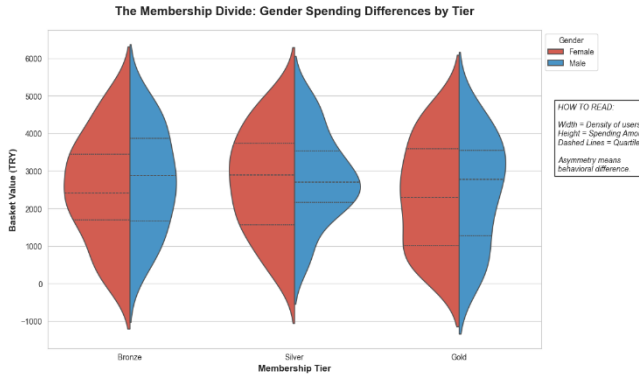

Figure 2: The Membership Divide: Gender Spending Differences by Tier (Split Violin Plot)

Fig. 2 uses a kind of graph to compare how much money females and males spend. The graph is split into two parts, one for females and one for males. Females are on the left side of each violin graph, and their part is red. Males are on the left side of each violin graph and their part is blue. This graph shows spending on bronze, silver and gold membership tiers. The width of each part of the graph shows how many customers spend money at levels. There are dashed lines inside each part that mark the quartiles for customer spending levels, for females and males. The chart illustrates a balanced distribution which means that whether you are a male or a female does not really affect how you spend money in these groups. Some important things that have been found include, when observed Uniform Median Spending, it can be seen that the lines for males and females are close to each other. This is true, for all levels of spending. It shows that the average man and the average woman spend about the amount of money each time they make a purchase. The Uniform Median Spending is a way to see this because it gives us a clear picture of how much people typically spend. The way people spend money is similar. When gold tier is considered, it is seen that both men and women tend to spend the money in the same range, which is between 2000 and 4000 Turkish Liras. The peaks of spending are the same, for both gold tier men and gold tier women.

Tier Progression: It is observed that people are spending money as we go from bronze to gold. The gold tier is interesting because people are spending a lot of money, sometimes up to 6000 TRY. This means that people who are gold tiers buy lots of things at once, which is different from the bronze tier where people are spending similar amounts of money. This absence of the significant gender gap suggests that marketing strategies for premium tiers should prioritize unified, high-value incentives rather than gender-segmented approaches.

B.3.) Discount Dynamics: How do discount rates vary across different membership tiers?

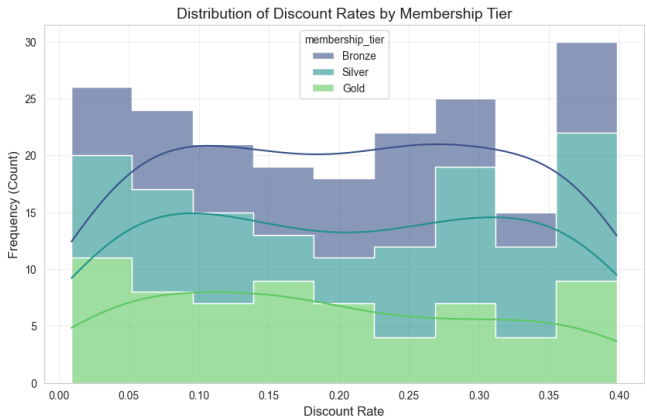

Figure 3: Distribution of Discount Rates by Membership Tier (Histogram)

In Fig. 3, a histogram analysis was conducted using the discount rate variable to examine the distribution of discount strategies on different customer membership tiers within the e-commerce platform. The analysis provides a comparative overview of the frequency and density of discount rates applied to customers in the bronze, silver, gold, and guest categories. Upon examining the graph, it is observed that the discount rates in the dataset do not follow a normal distribution, instead, they exhibit a multimodal structure with clusters at specific rates. The right part of the distribution indicates that lower discount rates (0.05 - 0.15 range) are applied more frequently, while higher discount rates (>0.30) are rarer and can be considered outliers, also. The following findings stand out when evaluating the differences between tiers.

1) Guest Segment: This group, which spans a wide range on the histogram and makes irregular peaks, suggests that the discount strategy for non-members is constructed in a variable manner, likely to focus on new customer acquisitions.

2) Gold and Silver Segments: The distribution curves for these loyalty program members are more consistent in specific ranges. Notably, the increased density of gold memberships in the right tail of the graph (high discount regions) confirms that benefits increase in a very small amount as the loyalty level rises.

3)Clustering Analysis: For all tiers, the kernel density estimate (KDE) curve demonstrates that discounts are not random, but they are likely clustered in specific percentiles, possibly tied to campaign periods or basket value thresholds. In conclusion, the relationship between membership tier and discount rate relates with the company's strategy of rewarding loyal customers. However, the high variance observed in the "Guest" segment points to a dynamic pricing policy aimed at measuring the price sensitivity of this group.

B.4.) The Value-Satisfaction Matrix: Are high-spending customers actually happy?

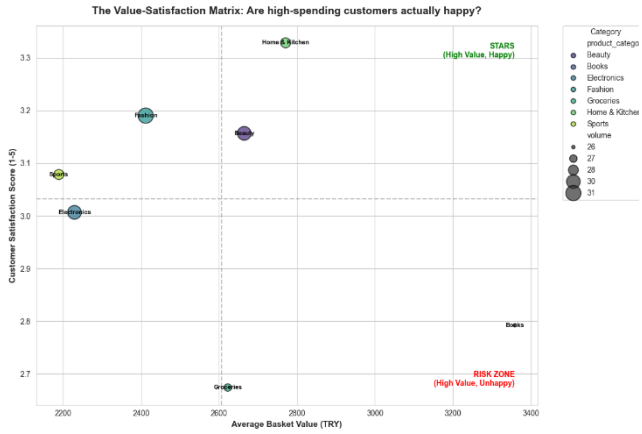

Figure 4: The Value-Satisfaction Matrix (Bubble Chart)

In Fig. 4, the analysis reveals how the average basket value and customer satisfaction are connected. The size of the bubbles shows the sales volume. The chart is divided into parts to help us understand if selling lots of stuff is efficient or not for how people think of the brand. The basket value and customer satisfaction needs to be taken into consideration in order to perceive this graph. The total sales volume is the main factor to consider. Some differences can be observed, in how things work in the main areas: The home and kitchen category seems to be robust, and it is one of the categories that are in the adequate spot. People spend lots of money on that category, and they seem to be happy with the product that they buy due to the satisfaction score it has, meaning home and kitchen is the outstanding category for business. The home and kitchen is the factor that bring in the main income; moreover people tend to give high satisfaction scores to that category so it can be considered as a vital part of the business. The "Risk Zone" is the area where "Groceries" is located, which is located at the right. People spend lots of money on groceries . They are not very happy with what they get. This is a problem because people are spending a lot, meaning they are not satisfied. It can be interpreted as the products are not optimal or the service is not compelling for the people. If something is not implemented to fix this, the people who brings in money for the company on groceries category might not come back to buy again. Therefore, it can be stated that "Risk Zone" is the primary concern for grocery category. The books category, on the other hand, exhibits a remarkably high average basket value but suffers from notably low customer satisfaction, placing it critically close to the risk zone.

V.) Conclusion

In this report, it was aimed to unveil the trends behind spending behavior among the users of this e-commerce business by analyzing the provided data set. By the findings of each research question several implementations can be made by the company. First, company can focus on grocery category to improve that area due to its low satisfaction score despite its high transaction value. Second, further membership opportunities can be considered to attract people to buy memberships. Additionally, adopting a new model for guest type for high-variance discounting strategy can be implemented to achieve overall efficiency. Finally, implementing new strategies to attract males which are tend purchase in beauty category can be taken into consideration by the company. Addressing these areas will ensure a more sustainable and customer-centric environment for the company.